

GROSSETO, Italy

WMO: 162060

Lat: 42.760N

Lon: 11.072E

Elev: 15

StdP: 14.69

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	28.7	31.7	13.2	10.8	42.6	18.0	13.6	42.3	33.6	48.6	29.5	47.1	8.4	40	0.484

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	22.4	94.0	70.7	91.3	70.5	88.8	70.1	78.2	84.6	76.5	83.3	74.9	82.2	13.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.5	138.1	82.2	74.6	129.4	80.6	72.6	120.9	79.3	41.7	84.6	39.9	83.4	38.3	82.3	87.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
28.7	23.9	20.5	DB	23.4	98.8	3.5	2.7	20.9	100.8	18.9	102.3	16.9	103.8	14.4	105.8
			WB	22.1	81.2	3.2	2.3	19.8	82.9	17.9	84.2	16.0	85.5	13.7	87.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	60.6	46.2	47.0	51.3	56.3	63.8	71.6	76.6	77.1	70.1	63.2	54.6	47.8
	DBStd	12.03	5.21	5.31	4.56	4.46	4.31	4.87	3.89	4.01	4.64	4.92	5.90	5.68
	HDD50	424	139	108	39	5	0	0	0	0	0	1	25	107
	HDD65	2797	582	504	424	262	74	6	0	0	10	90	313	532
	CDD50	4282	22	24	80	195	428	649	825	842	603	411	163	40
	CDD65	1180	0	0	0	1	37	205	360	377	163	36	1	0
	CDH74	12014	0	0	1	21	404	2129	4067	4000	1238	152	2	0
	CDH80	4739	0	0	0	1	72	757	1813	1794	295	7	0	0
Wind	WSAvg	8.5	8.6	9.1	9.4	8.6	7.6	7.9	8.2	7.9	8.5	8.2	8.9	9.0
Precipitation	PrecAvg	25.7	2.3	2.0	2.1	1.9	1.5	1.1	0.6	1.3	2.8	3.2	4.0	2.6
	PrecMax	36.1	4.7	4.9	6.3	4.9	3.1	3.2	3.1	5.7	10.6	8.2	14.0	6.9
	PrecMin	16.7	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.4	0.0
	PrecStd	5.4	1.6	1.2	1.5	1.1	0.8	0.9	0.7	1.5	2.3	2.1	3.2	1.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.4	64.1	70.4	76.8	86.0	95.3	96.5	98.6	89.6	80.7	71.1	64.3
		MCWB	56.0	53.1	56.2	60.1	65.6	70.1	70.7	70.9	67.8	65.8	62.9	58.3
	2%	DB	59.3	61.0	66.3	73.1	81.8	90.4	93.4	94.9	86.1	77.6	68.2	61.2
		MCWB	54.2	52.6	54.5	58.8	64.1	68.6	70.2	71.1	69.0	66.8	61.8	56.2
	5%	DB	57.6	59.1	64.0	70.0	78.4	87.5	91.2	91.5	83.5	75.4	66.3	59.3
		MCWB	52.7	52.3	54.3	58.2	63.4	68.2	70.4	71.4	68.2	66.2	60.7	54.5
	10%	DB	55.7	57.2	61.3	67.1	75.4	84.4	88.9	88.4	80.8	73.1	64.3	57.4
		MCWB	51.2	51.4	53.4	57.4	62.8	68.0	70.2	71.2	67.7	65.3	59.2	53.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.8	57.1	60.3	64.8	71.5	77.6	80.6	81.9	78.0	72.4	66.5	59.9
		MCDB	60.4	60.3	64.9	71.4	78.2	85.0	86.7	87.7	82.2	76.0	69.9	62.7
	2%	WB	55.8	55.4	58.2	62.3	68.8	74.7	78.1	78.5	75.3	70.3	63.6	57.6
		MCDB	58.4	58.8	62.8	68.5	75.6	82.4	85.0	84.7	80.0	74.4	66.7	60.4
	5%	WB	54.1	53.9	56.7	60.5	66.7	72.7	76.2	76.8	73.1	68.6	61.7	55.7
		MCDB	56.9	57.5	61.2	66.5	73.8	80.9	83.6	83.4	78.7	73.0	65.2	58.5
	10%	WB	52.1	52.3	55.1	58.8	64.7	70.8	74.5	75.1	70.9	66.7	59.6	53.7
		MCDB	55.2	56.1	59.6	64.8	72.2	79.3	82.6	82.4	77.3	71.2	63.4	57.0
Mean Daily Temperature Range	5% DB	MDBR	16.3	18.1	19.2	20.2	21.1	22.5	22.8	22.4	20.2	18.1	15.6	15.6
		MCDBR	18.0	20.4	23.5	24.5	25.8	26.5	25.9	26.1	22.7	20.4	16.1	17.0
		MCWBR	13.9	14.9	15.2	14.5	13.0	11.3	10.8	11.2	11.7	12.8	11.8	13.0
	5% WB	MCDBR	15.6	17.5	19.1	20.5	22.3	22.1	21.5	21.5	18.4	16.9	14.0	14.9
		MCWBR	13.5	14.3	14.7	14.6	14.7	13.0	13.2	13.2	13.0	13.0	11.6	12.9
Clear-Sky Solar Irradiance	taub	0.339	0.358	0.392	0.425	0.431	0.429	0.421	0.419	0.406	0.393	0.369	0.338	
	taud	2.458	2.387	2.298	2.223	2.251	2.302	2.326	2.348	2.370	2.389	2.392	2.467	
	Ebn at Noon	252	265	268	267	268	269	269	266	261	248	236	240	
	Edh at Noon	24	30	37	43	43	41	39	37	34	30	26	22	
All-Sky Solar Radiation	RadAvg	534	826	1216	1632	2006	2284	2335	2028	1497	977	591	473	
	RadStd	54	84	143	119	160	111	106	102	76	86	55	51	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.88	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+245	+130
(47)	Regional (25 neighbors)	+0.59	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+130	N/A

Nomenclature: See separate page